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Global molecular landscape of early MASLD progression in obesity

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eLife Assessment

The authors provide a **useful** integrated analytical approach to investigating MASLD focused on diverse multiomic integration methods. The strength of evidence for this new resource is **solid**, as analyses highlight the importance of previously-described pathophysiologic processes, as well as unveil several new mechanisms as key features of MASLD in obese patients.

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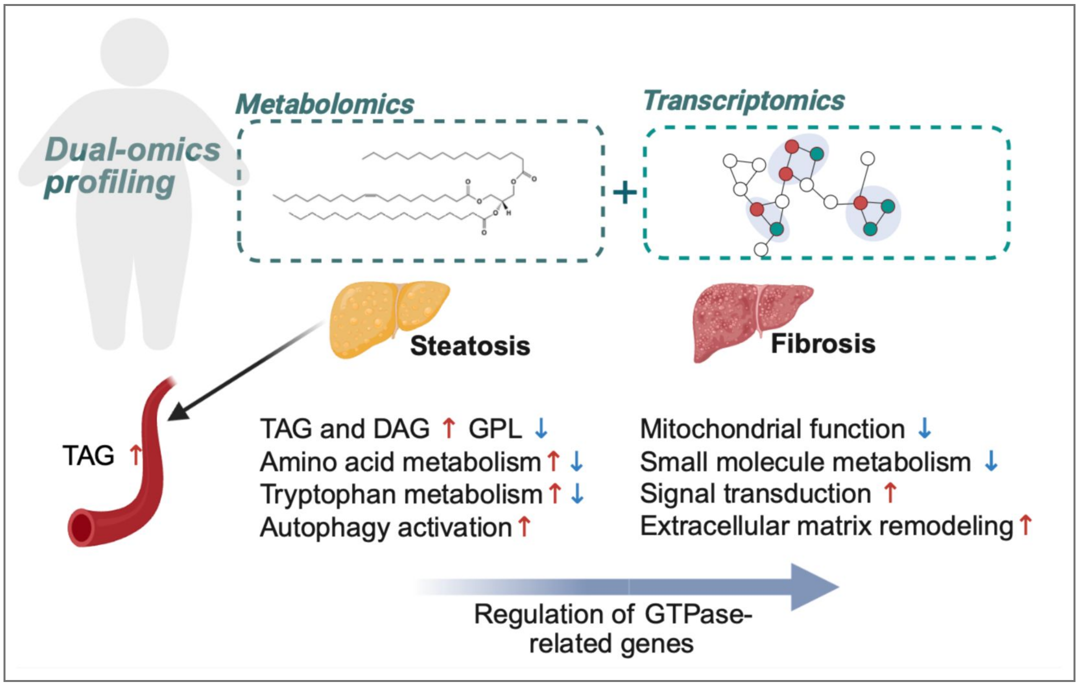
Abstract

Metabolic dysfunction-associated steatotic liver disease (MASLD) is often asymptomatic early on but can progress to irreversible conditions like cirrhosis. Due to limited access to human liver biopsies, systematic and integrative molecular resources remain scarce. In this study, we performed transcriptomic analyses on liver and metabolomic analyses on liver and plasma samples from morbidly obese individuals without liver pathology or at early-stage MASLD. While, the plasma metabolomic profile did not fully mirror liver histological features, dual-omics integration of liver samples revealed significantly remodeled lipid and amino acid metabolism pathways. Integrative network analysis uncoupled metabolic remodeling and gene expression as independent features of hepatic steatosis and fibrosis progression, respectively. Notably, GTPases and their regulators emerged as a novel class of genes linked to early liver fibrosis. This study offers a detailed molecular landscape of early MASLD in obesity and highlights potential targets of obesity-linked liver fibrosis.

Graphical abstract



Introduction

Metabolic dysfunction-associated steatotic liver disease (MASLD) is the most common chronic liver disease, affecting over 30% of adults worldwide [1]. The prevalence of MASLD is closely associated with the epidemic of obesity, with an estimated 75% prevalence among individuals with obesity compared to 32% in the general population [2]. As liver manifestation of the metabolic syndrome, MASLD can cause, and is also caused by concurrent metabolic abnormalities commonly seen in obese individuals, such as insulin resistance, diabetes, dyslipidemia, and hypertension [3, 4].

Early stages of MASLD are characterized by hepatic steatosis, which is the excessive deposition of triacylglycerol (TAG) rich lipid droplets (LD) within the liver parenchyma and is usually reversible. However, in a subset of patients, the disease can develop into inflammatory and ballooning stages termed metabolic dysfunction-associated steatohepatitis (MASH), a more severe form of the disease with an increased incidence and severity of fibrosis. Fibrosis can be present at any stage of the MASLD spectrum of disease and at each stage is increasingly associated with treatment complications, liver-related and overall mortality [5–7]. Unresolved fibrosis can progress to end-stage diseases including hepatic cirrhosis, and hepatocellular carcinoma, lethal malignancies with limited treatment options [8, 9].

Metabolic dysfunction is a primary feature and contributor of MASLD pathogenesis [10]. Hepatic steatosis is often accompanied by increased glucose production [11], elevated de novo lipogenesis [12], and disrupted cholesterol homeostasis [13]. Mitochondrial respiration is adaptively upregulated to prevent lipid accumulation in obesity [14], but with MASLD progression, mitochondria can become dysfunctional and exacerbate metabolic dysregulation [15]. These interconnected metabolic phenotypes form the basis for the development of MASLD [16, 17]. Recent advancements in metabolomics have greatly enhanced the understanding of the molecular systems biology of MASLD [18, 19], however, few studies have combined such analysis with transcriptomics to provide integrative insights into disease pathogenesis in humans.

Fibrosis is the only histological feature of MASLD associated with liver-related mortality and morbidity [20]. In the injured liver, damaged and apoptotic hepatocytes modulate the crosstalk between hepatocytes, liver macrophages (Kupffer cells), and hepatic stellate cells (HSCs) by releasing fibrogenic cytokines (e.g., TGF-β) and activating multiple signaling pathways [21, 22]. As a result of the multicellular response, HSCs transdifferentiate into active, collagen-producing myofibroblasts, driving fibrogenesis and the excessive deposition of extracellular matrix (ECM) [22]. Since fibrosis is a key prognostic marker of MASLD progression, the shift from steatosis to fibrosis marks a critical point in the disease, offering a key opportunity for intervention to prevent further progression [23]. Despite the availability of certain guidelines for clinical management of MASLD [24, 25], the molecular and metabolic changes underpinning the key transitions driving disease progression in the context of obesity remain poorly understood. This poses a challenge to early disease management and prevention.

In this study, we generated a comprehensive map of gene expression and metabolomic profiles to delineate the molecular events associated with early-stage MASLD progression in obesity. Using this multi-omic resource, we focused our investigation on key molecular changes associated with (a) the transition from obesity with normal hepatic histology to MASLD and (b) the onset of liver fibrosis. Our data reveal distinct molecular signatures underlying steatosis and fibrosis progression, offering a detailed molecular portrait of the liver in early MASLD and highlighting potential therapeutic targets for reversing fibrosis at initial stages.

Results

Overview of the study

We analyzed samples from 109 obese individuals recruited before bariatric surgery at The Avenue, Cabrini, or Alfred Hospitals in Melbourne, Australia (Table 1 and Table S1). Following the exclusion criteria described in Materials and Methods, 33 individuals lacked histological abnormalities (no MASLD) and 76 had MASLD (Figure 1A). Notably, 83 individuals (76%) were females in this cohort. Most individuals with obesity were in the early disease stages, with 74 individuals (68%) displaying grade 0-1 steatosis and 83 (76%) grade 0-1 fibrosis. Hepatic inflammation and ballooning were mild, with 9 cases exhibiting grade 2 or higher inflammation and only one case showing grade 2 ballooning (Figure 1B and Table 1). In clinical tests, MASLD patients displayed worse liver function (higher alanine transaminase (ALT)/aspartate aminotransferase (AST) ratio and gamma-glutamyl transpeptidase (GGT) levels), higher non-high-density (HDL) lipoprotein cholesterol and blood triglyceride levels, higher C-peptide, and insulin resistance compared to those with ‘No MASLD’ (Table 1, $p < 0.05$), confirming correct classification. Liver fibrosis was strongly correlated with insulin resistance, while steatosis grades were most correlated with the levels of plasma lipids (Figure 1C).

We performed parallel transcriptomic profiling of liver slices and metabolomic analysis of liver and plasma samples (see Materials and Methods). Untargeted metabolomics (covering polar & non-polar metabolites including lipids) enabled semi-quantification of 1104 liver and 962 plasma metabolites, respectively (Figure 1D). Plasma urea and creatinine levels from untargeted metabolomics closely correlated with clinical assays (Figure S1A), supporting data reliability. Principal component analysis (PCA) revealed distinct clustering of the liver metabolome by MASLD status, but not in plasma (Figure 1E). The liver transcriptome was modestly separated between individuals with and without MASLD along the main principal components (Figure 1F), indicating that transcriptional programs may be shaped by both MASLD histological progression and confounding metabolic and biological processes in obese individuals (Table S2).

As steatosis and fibrosis are the key histological features of MASLD, our statistical analyses focused on identifying molecular determinants linearly associated with their progression (Table S3-5). Models were adjusted for patient characteristics (age, sex, and BMI) and type 2 diabetes, the latter due to its differing prevalence across disease groups (Table 1). Moreover, we compared our transcriptomic data with those from two additional published cohorts: the Virginia cohort (VA cohort, GSE130970) [26] with a disease spectrum comparable to that of our study, and the European cohort (EU cohort, GSE135251) [27, 28] with a broader spectrum of the disease covering advanced MASLD pathologies (Figure 1G).

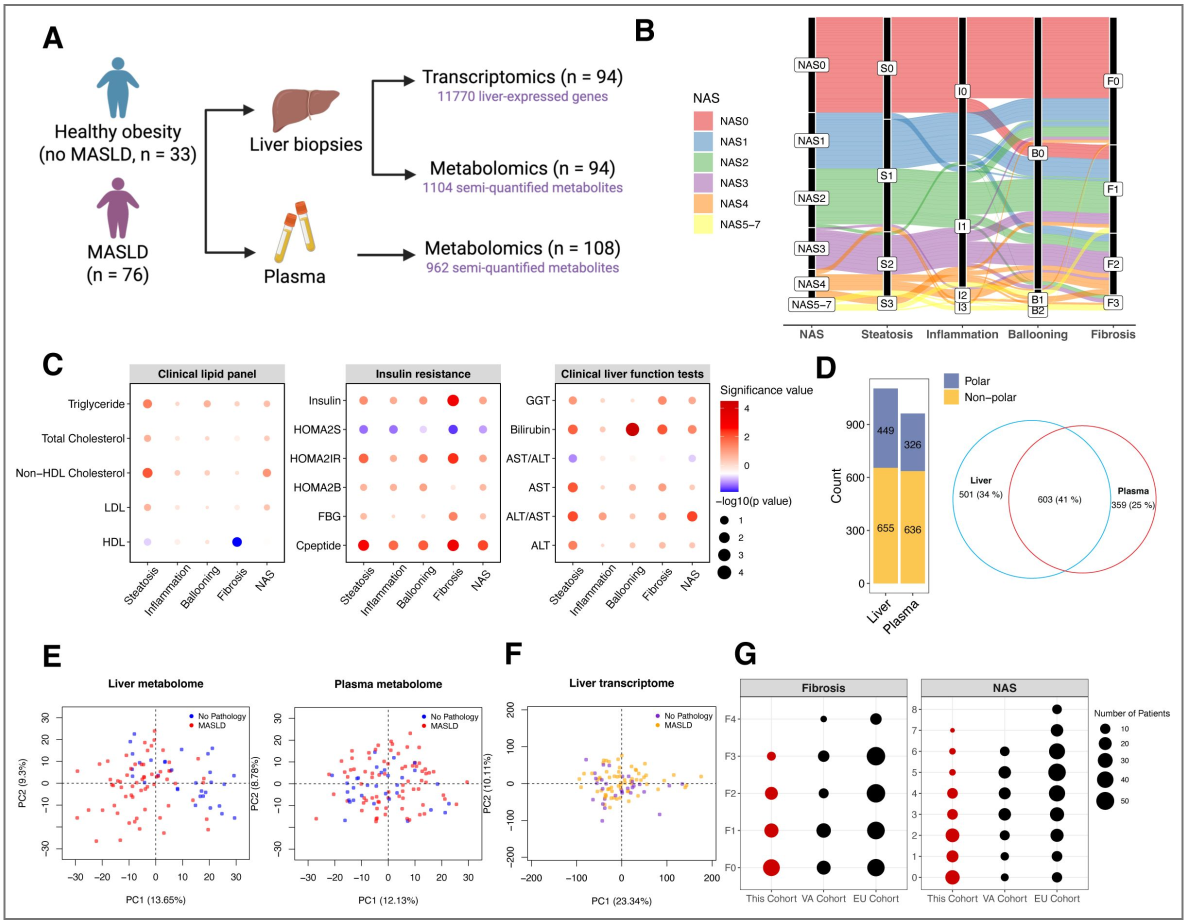


Figure 1. **Study overview.**

(A) The overall study design. (B) Alluvial diagram of patient composition of groupings across liver histological features. (C) Relationship between clinical test results and stages of liver histological features. Node size was determined by −log10(p value) in ANOVA tests. Color indicates the significance degree [−log10(p value)] and the direction of change from early to late stages. (D) Number of metabolites analyzed in the liver and plasma. (E-F) Principal component analysis of liver metabolome, plasma metabolome, and liver transcriptome. (G) Disease spectra covered in this cohort and comparison to two published datasets.

	Patients (N)	No MASLD (N = 33)	MASLD (N = 76)	P value
Patient information				
Age (years), median ± MAD	109	36 (14.8)	41.5 (9.6)	0.942
Sex (male), n (%)	108	4 (12.1)	22 (29.3)	0.092
Diabetes, n (%)	109	2 (6.1)	27 (35.5)	0.003
Hypertension, n (%)	93	6 (18.2)	16 (26.7)	0.505
BMI (kg/m2), median ± MAD	108	42.6 (6.6)	44.5 (9.1)	0.045
Clinical chemistry parameters				
ALT (U/L), median ± MAD	104	24 (11.9)	42 (25.2)	0.507
AST (U/L), median ± MAD	104	24 (7.4)	30 (13.3)	0.39
ALT/AST, median ± MAD	104	1 (0.3)	1.33 (0.4)	0.005
GGT (U/L), median ± MAD	104	23 (10.4)	28 (13.3)	0.041
Total cholesterol (mmol/L), median ± MAD	104	4 (0.9)	4.15 (1.0)	0.072
HDL (mmol/L), median ± MAD	102	1.04 (0.2)	0.94 (0.2)	0.053
LDL (mmol/L), median ± MAD	101	2.2 (0.7)	2.6 (0.7)	0.081
Non-HDL cholesterol (mmol/L), median ± MAD	108	2.76 (0.9)	3.27 (1.0)	0.003
Blood triglyceride (mmol/L), median ± MAD	103	1.2 (0.6)	1.5 (0.6)	0.013
Insulin (mU/L), median ± MAD	100	7.95 (5.6)	10.35 (7.1)	0.043
FBG (mmol/L), median ± MAD	102	4.9 (0.7)	5.2 (1.0)	0.114
C-peptide (nmol/L), median ± MAD	101	0.8 (0.4)	1.12 (0.4)	0.002
HOMA2–IR, median ± MAD	92	1.13 (0.7)	1.43 (0.9)	0.034
Liver histology				
Steatosis (S0/S1/S2/S3)	109	33/0/0/0	0/41/27/8	<0.001
Inflammation (I0/I1/I2/I3)	109	30/3/0/0	21/46/7/2	<0.001
Ballooning (B0/B1/B2)	109	33/0/0	65/10/1	0.07
NAS (N0/N1/N2/N3/ N4/N5/N6/N7)	109	30/3/0/0/ 0/0/0/0	0/15/29/13/ 14/2/2/1	<0.001
Fibrosis (F0/F1/F2/F3)	109	27/6/0/0	23/27/20/6	<0.001

MASLD: Metabolic dysfunction-associated steatotic liver disease; Median absolute deviation; BMI: Body mass index; ALT: Alanine transaminase; AST: Aspartate aminotransferase; GGT: Gamma-glutamyl transpeptidase; HOMA2 – IR: Homeostasis model assessment 2 of insulin resistance; HDL: High-density lipoprotein; LDL: Low-density lipoprotein; NAS: Nonalcoholic fatty liver disease activity score.

Table 1. **Patient characteristics**

Global investigation of the metabolome in obese MASLD patients

As expected, the liver metabolome was extensively remodeled in individuals with MASLD. Steatosis is the primary histological feature linked to liver metabolome, with 206 metabolites showing significant positive associations and 242 showing negative associations ($q < 0.05$, Figure 2A and Figure S1B). For example, we observed higher levels of glycerolipids (GLs, e.g. TAGs, diacylglycerols [DAGs]) and lower levels of membrane lipid classes, especially glycerophospholipids (GPLs) (Figure 2A-B and Figure S1B). Specifically, TAGs with 0 to 5 double bonds in the fatty acyl chains were positively associated with the histological outcomes, whereas TAGs containing at least one polyunsaturated fatty acid (PUFA) chain (e.g. number of double bonds > 5) were negatively associated with disease progression, particularly with steatosis (Figure S1C). Our findings are likely the results of increased de novo lipogenesis and elevated hepatotoxicity associated with saturated fatty acids [29, 30].

The plasma metabolome displayed limited associations with the histological features (Figure 2A), with statistically significant associations primarily with steatosis, such as TAGs (q value < 0.05, n = 22) and ether-linked TAG (TAG-O, q value < 0.05, n = 8) species. To better interpret the plasma metabolome data, we performed partial correlation network analysis to assess the associations among circulating metabolites, clinical variables, and hepatic histology in individuals with obesity [31]. The network demonstrates the highest connectivity of plasma metabolites with clinical variables and fewer connections with hepatic histology features (Figure 2C). This implies that plasma metabolome largely reflects other metabolic conditions such as kidney function, dyslipidemia, and insulin resistance rather than hepatic features. Most plasma lipid classes were correlated with blood lipoprotein cholesterol levels (Figure 2D). However, in the context of obesity, blood lipoprotein levels did not show strong correlation with MASLD progression (Figure S2), indicating that their broad impact on plasma metabolome may mask the signals from hepatic abnormalities.

We next compared steatosis- and fibrosis-associated metabolite changes in the liver and plasma (Figure 2E). Plasma GLs exhibited similar but weaker associations with liver steatosis compared to hepatic GLs, whereas the depletion of PUFA-containing TAGs in the liver was not mirrored in circulation. Although plasma polar metabolites showed changes in the context of steatosis, potential circulating markers emerged for liver fibrosis with mild statistical significance. With fibrosis progression, there was a trend toward increased levels of tyrosine, quinolinic acid, lactic acid, and kynurenic acid in plasma ($p < 0.05$, $q > 0.05$) (Table S3), consistent with their reported roles as key hepatic metabolites implicated in MASLD progression and as proposed circulating biomarkers for MASLD severity or biopsy-proven MASH [32–35]. Overall, despite elevated TAG levels and potential fibrosis indicators, the circulating metabolome is less indicative of early-stage MASLD-related alterations in individuals with obesity and obesity-related co-morbidities.

Integrative view of liver metabolism remodeling

We next integrated the hepatic metabolome and transcriptome to characterize metabolic remodeling in obese individuals during early disease development.

Lipid metabolism

In the liver, accumulation of GLs and mild reduction of GPLs were the main features of the metabolome changes (see Figure 2). Consistent with this observation, we identified genes implicated in the homeostasis of GLs and GPLs (Figure 3A and Figure S3). Genes such as *DGAT2*, *PNPLA3*, and *PLIN3* play a role in LD formation and TAG and DAG metabolism [36]. GPL metabolism was also markedly altered, including key genes such as *LPCAT1*, *PLD2*, *PCYT2*, *ETNK*2 and that are implicated in a compensatory response to the shift in phospholipid metabolism and the increased turnover of GPLs [37, 38]. In individuals with advanced hepatic fibrosis, expression levels of the genes involved in primary bile acid biosynthesis, such as *SLC27A5*, *AMACR*, *ACOX2*, *AKR1C4*, and *BAAT*, were lower. In addition, a group of cytochrome P450 (CYP) genes were downregulated during the progression of fibrosis, particularly those involved in CYP-dependent PUFA metabolism (i.e., linoleic acids and arachidonic acids into bioactive molecules) and steroid hormone

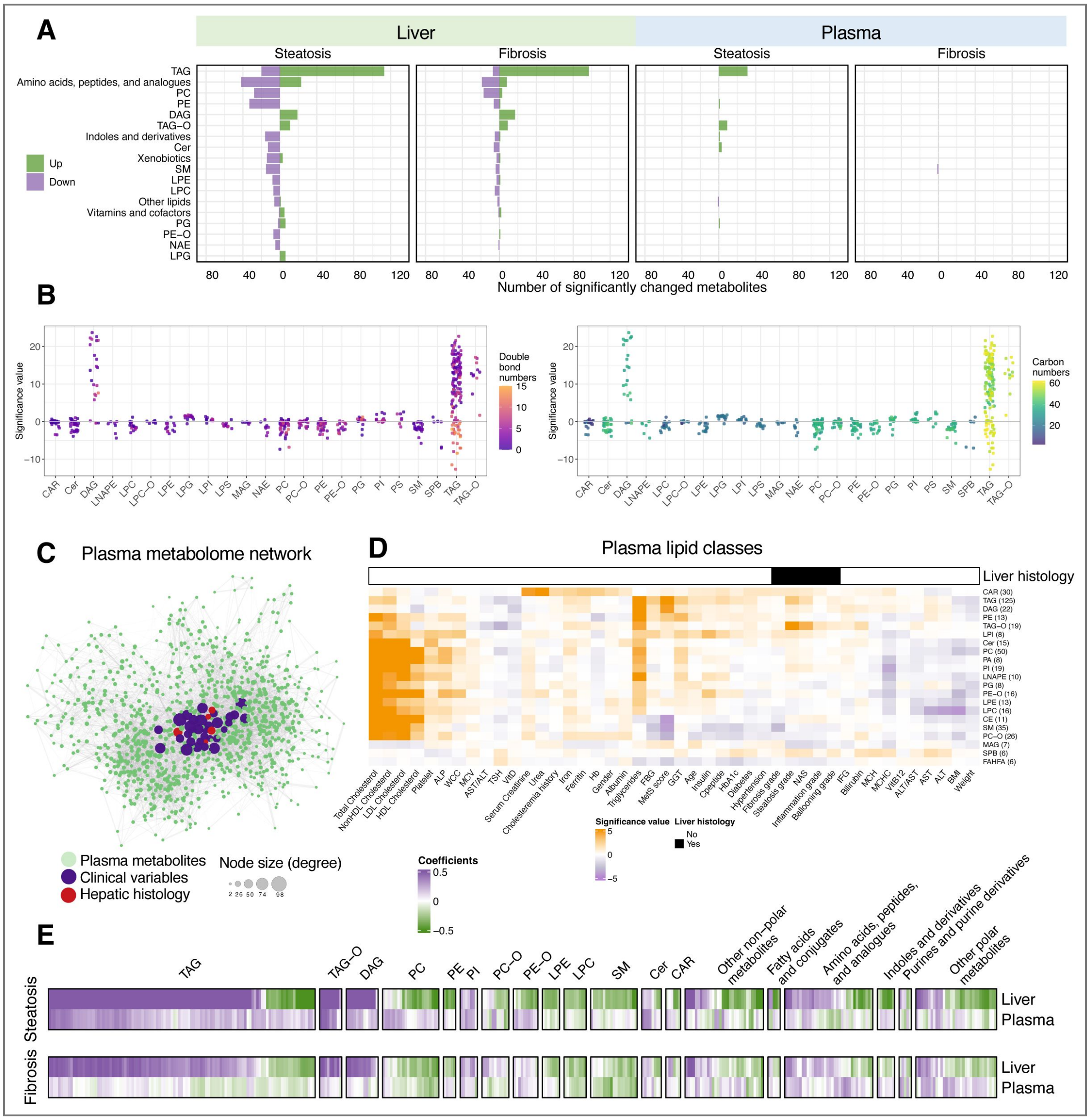


Figure 2. **Hepatic and circulating metabolome in obese individuals with MASLD.**

(A) Number of metabolites in each class that were significantly associated with histological features in the liver (q value < 0.05). Metabolite classes with at least 5 metabolites with significance are shown in the plot. (B) Associations between steatosis grades and lipid species in each lipid class. Dots were colored by double bond numbers (left) and carbon numbers (right) of lipid species, respectively. (C) Partial correlation network of the plasma metabolome and clinical covariates. Node size reflects the degree of connectivity, with larger nodes indicating connections to a greater number of metabolite nodes. (D) Heatmaps for pairwise analysis between plasma lipid classes and clinical variables. Linear regression analysis was performed for numerical variables, while logistic regression was conducted for binary variables. Significance values refer to –log10(p value)*sign(coefficients) from regression models. (E) Comparisons of liver metabolome and plasma metabolome regarding their associations with liver steatosis and fibrosis.

biosynthesis [39, 40] (Figure 3A). Overall, our data highlights a number of genes encoding the enzyme subunits of anabolic and catabolic lipid metabolism in early MASLD, and some of these gene signals were consistently observed in both the EU and VA cohorts [26, 27].

Dysregulated metabolic pathways during steatosis progression

To explore the variation in metabolic activities using dual-omic descriptors, we prioritized key pathways with both dysregulated metabolites and gene expressions (Figure 3B). In addition to significant GL and GPL remodeling, we also identified altered pathways of amino acid metabolism at both omics levels. Elevated levels of amino acids including glycine, glutamic acid, arginine, serine, and alanine in steatotic livers may reflect increased collagen synthesis and ECM remodeling during MASLD progression [41, 42] (Figure 3C-D). Notably, aromatic amino acids including phenylalanine, tyrosine, and tryptophan were lower in subjects with advanced steatosis (Figure 3D). Dual-omics data revealed consistent downregulation of tryptophan metabolism, encompassing both the indole and melatonin pathways (Figure 3C). However, quinolinic acid, a key metabolite in the kynurenine pathway and a product of tryptophan catabolism, was significantly elevated in association with both hepatic steatosis and fibrosis. Collectively, early MASLD development is associated with reduced aromatic amino acid levels and the altered tryptophan catabolic flux in the liver, potentially reflecting the aberrant gut–liver axis communication in obese MASLD patients [43–45].

Moreover, homocysteine and its upstream metabolite S-adenosylhomocysteine levels were significantly higher in steatotic livers, indicating dysregulated methylation with excessive lipid storage [46]. However, the accumulation these metabolites was mitigated by fibrosis-associated downregulation in key enzymes including *GNMT*, *MAT1A*, and *AHCY* (Figure 3C), which have been linked to liver pathogenesis in in vivo models [47–50]. These suggest that an imbalance in methylation reactions of DNA and proteins may be an important mediator of liver fibrogenesis in obese individuals.

In addition, we observed elevated levels of the antioxidant coenzyme Q10 (CoQ10; ubiquinone) and its intermediates CoQ6 and CoQ9 in the steatotic liver (Figure 3D). Higher levels of two other antioxidants, taurine and alpha-tocopherol (vitamin E), were also detected. Another indicator of oxidative stress, hydroxybutyric acid, was increased in advanced steatosis and is also recognized as an early marker of insulin resistance [51]. Overall, the progression of liver steatosis is accompanied by extensive changes in amino acid metabolism and oxidative stress regulation.

Dysregulated genes involved in mitochondrial function and autophagy

In the liver metabolome, lower levels of hepatic long-chain acyl-carnitine (CAR) species were observed with advanced steatosis (CAR 18:2 and 20:4, Table S4), suggesting dysregulated fatty acid transmembrane transport and β-oxidation [52, 53]. To systematically assess mitochondrial functions beyond β-oxidation, we mapped steatosis and fibrosis-associated genes to MitoCarta3.0 [54]. Distinct mitochondrial dysfunction patterns emerged: steatosis involved 26 downregulated and 55 upregulated mitochondrial-function genes, including 16 related to mtDNA maintenance, while fibrosis was linked to 151 downregulated (dark green) and only 15 upregulated (orange) genes (Figure 4A and Table S6). These transcriptional shifts suggest that hepatic fibrosis involves broad mitochondrial impairment, contributing to oxidative stress and reduced ATP production, which could favor the exacerbation of steatohepatitis [15].

Autophagy serves as a critical cellular response mechanism to the overload of intrahepatic TAG and cholesterol [55, 56]. Metabolomic analysis showed increased free cholesterol and reduced levels of CE 18:1 and 18:2, the predominant hepatic CE species [57] (Figure S4A), supporting prior findings that CE de-esterification contributes to elevated free cholesterol in MASLD [58]. Although autophagy has been linked to HSC activation [59, 60], its involvement in MASLD progression remains unclear. Using a list of predictive genes in autophagy activation [61], we found that autophagy activation and lysosome biogenesis were more strongly associated with steatosis than fibrosis, with upregulated key autophagy genes such as *ULK1* and *MTOR* (Figure 4B). Further investigation of over 600 autophagy genes [61] (Table S7 and Figure S4B) revealed active autophagy initiation in early MASLD, where 119 genes across pathways including mTORC and upstream effectors, lysosome, and

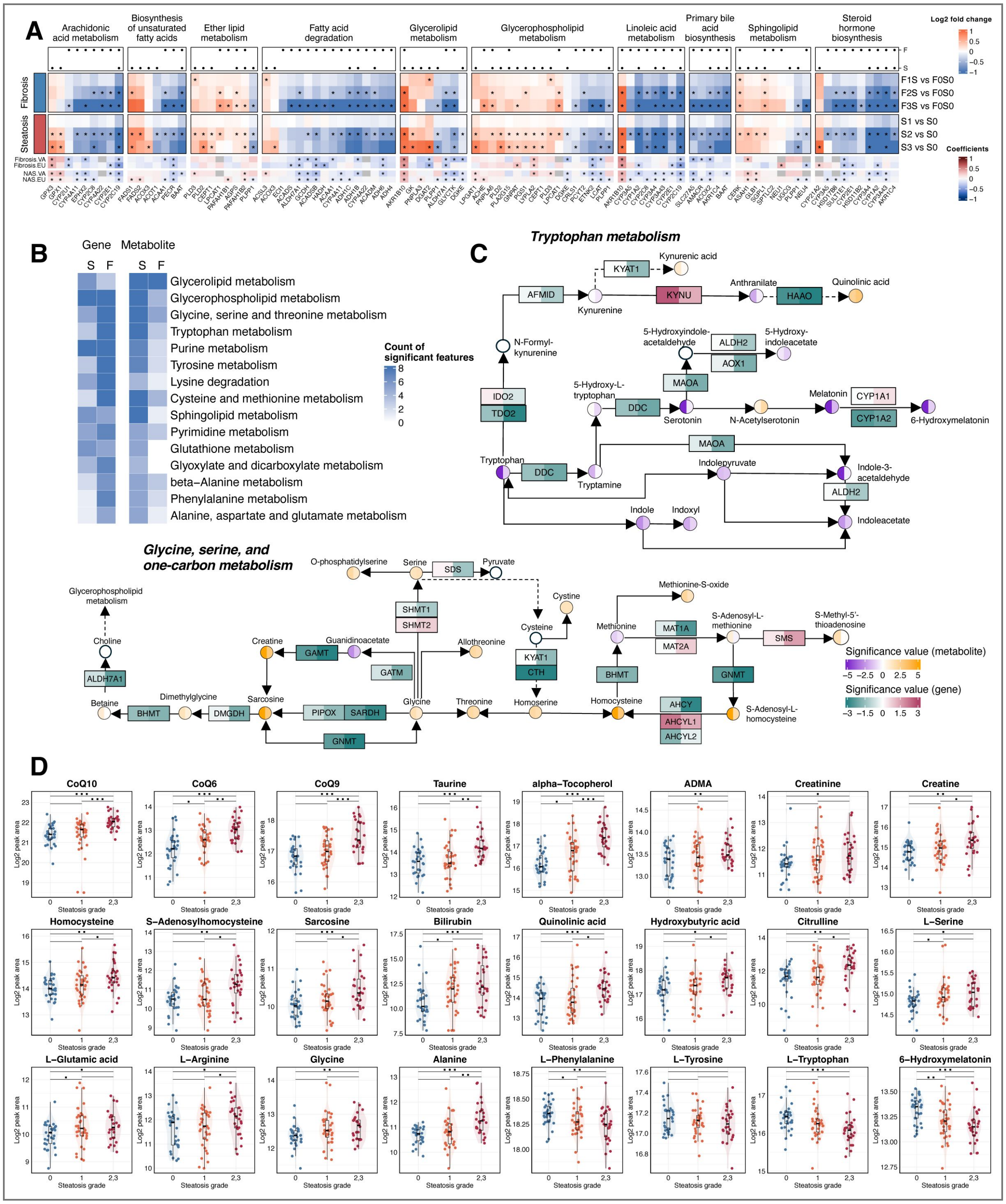


Figure 3. **Integrative view of key metabolic pathways implicated in liver metabolism in obese individuals with MASLD.**

(A) Heatmap of log2 fold changes from pair-wise analysis of lipid metabolism-related genes. Association with steatosis (S) and fibrosis (F) is indicated by black dots (q value < 0.05, top). Results were cross-referenced with two published cohorts (VA cohort, GSE130970; EU cohort, GSE135251, bottom). Asterisks indicate genes with a q-value < 0.05 and a consistent change in direction within our cohort. (B) KEGG metabolic pathways with at least 4 genes or metabolites significantly associated with steatosis [S] or fibrosis [F]. (C) Integrative map of gene and metabolite alterations associated with steatosis (left half of each box/circle) and fibrosis (right half of each box/circle). (D) Metabolite alterations corresponding to the advancement of steatosis grades (x-axis).

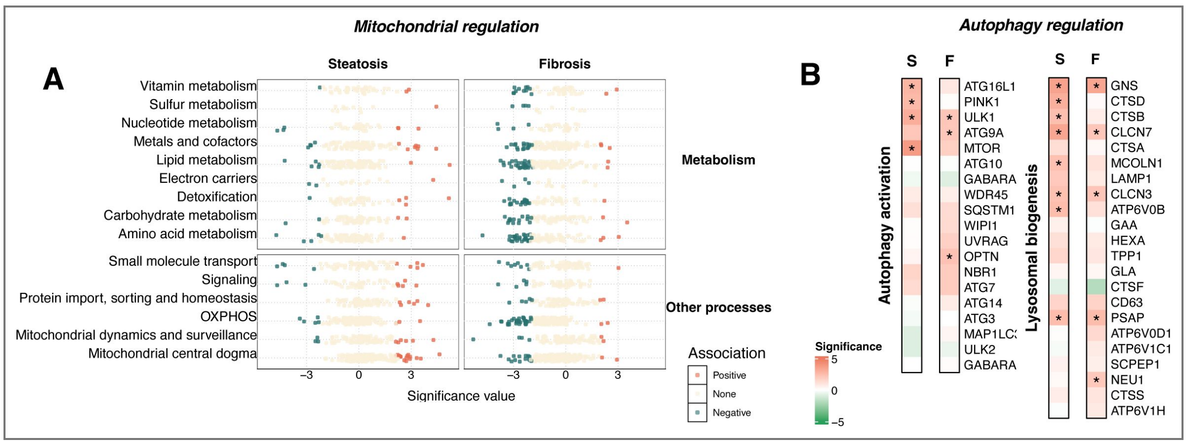


Figure 4. **Dysregulated mitochondrial function and autophagy during the progression of steatosis and fibrosis.**

Gene expression patterns related to mitochondrial metabolism (A), mitochondria-related biological processes (A), autophagy activation (B), and lysosomal biogenesis (B) with statistical association to steatosis and fibrosis. S, steatosis. F, fibrosis.

autophagy core assembly genes displayed altered expression levels. This suggests that autophagy serves as an adaptive response to hepatic lipotoxicity during steatosis, but its activation may decline as the disease progresses, as evidenced by the downregulation of key regulators such as *NR1H4*, *SIRT5*, *FOS*, and *EGR1* (Figure S4B), thereby impairing liver metabolism [62].

Molecular signatures of hepatic steatosis and fibrosis are mutually independent

To gain insights into the dual-omic data in the liver further, we performed a partial-correlation network analysis to integrate liver transcriptomic, metabolomic, and clinical data [31]. The resulting subnetwork highlighted distinct molecular signatures correlated with steatosis and fibrosis (Figure 5A). Steatosis was primarily associated with hepatic neutral lipid accumulation and related metabolomic alterations, whereas fibrosis was predominantly linked to transcriptional changes, including the upregulation of genes involved in ECM remodeling (e.g., *TGFB3*, *FSTL1*), signal transduction (e.g., *RHOU*, *DOK3*), and metabolism (e.g., *CYP2C19*, *PNPLA4*, *SGPL1*). These findings underscore the presence of distinct molecular pathways driving the progression of steatosis and fibrosis.

In our transcriptomics analysis, 992 and 1,251 gene markers were identified as significantly associated with the histological features of steatosis and fibrosis, respectively, with limited overlap between the two sets (Figure 5B and Table S5, q value < 0.05), suggesting unique transcriptomic regulations underlie each feature. Functional enrichment analysis further supported this distinction, revealing largely non-overlapping biological processes associated with each gene set (Figure S5 and Table S8).

To distinguish gene signatures uniquely linked to steatosis or fibrosis, we included the other histological feature as a covariate in the regression model. This allowed us to define steatosis-specific genes (independent of fibrosis) and fibrosis-specific genes (independent of steatosis) (Table S9). Since steatosis precedes fibrosis in the pathophysiology of MASLD, enrichment analysis of steatosis- and fibrosis-specific genes revealed a ‘pseudo-temporal’ progression of biological processes (Figure 5C). More pathways were found to be associated with fibrosis than with steatosis. Steatosis-specific upregulated genes were enriched in protein glycosylation, inflammatory response, lipid catabolism, and lysosome organization.

In contrast, fibrosis-specific genes showed downregulation of various metabolic pathways and upregulation of processes related to ECM remodeling, hypoxia, signaling, and cell morphogenesis. Additionally, apoptosis-related pathways were suppressed in steatosis but activated in fibrosis, while nucleotide metabolism showed the opposite trend, suggesting dynamic regulation of cell death and proliferation during MASLD development. These findings implicate distinct and lesion-specific gene regulatory programs in early MASLD progression.

Moreover, consistent with previous observations showing that hepatic fibrosis correlates with insulin resistance parameters in clinical assays (Figure 1C), we found that individuals with diabetic MASLD exhibited a greater number of downregulated genes as fibrosis progressed than non-diabetic MASLD individuals (Table S10). Since most of the suppressed genes in the diabetic subgroup are involved in metabolism (e.g., *BAAT*, *G6PC1*, *SULT2A1*, *MAT1A*), we hypothesize that diabetes may exacerbate the metabolic dysfunction associated with hepatic fibrosis progression.

Gene signatures of fibrosis initiation

To further explore representative gene signatures in the development of fibrosis, we identified 213 genes as progressive markers by comparisons of gene expression levels against two different baselines (no MASLD and steatosis; details provided in the methods; Table S11). Among them, 75 of these markers overlapped with fibrosis-associated genes in the VA cohort and 35 overlapped with the EU cohort (Figure 6A) [26], resulting in 130 novel fibrosis markers. Pathway analysis of these fibrosis markers highlighted prominent roles for signal transduction and ECM organization/disassembly, as expected (Figure S6A). To infer the cell type-of-origin for signals from bulk sequencing, we mapped progressive fibrosis markers onto a human liver single-cell atlas [63].

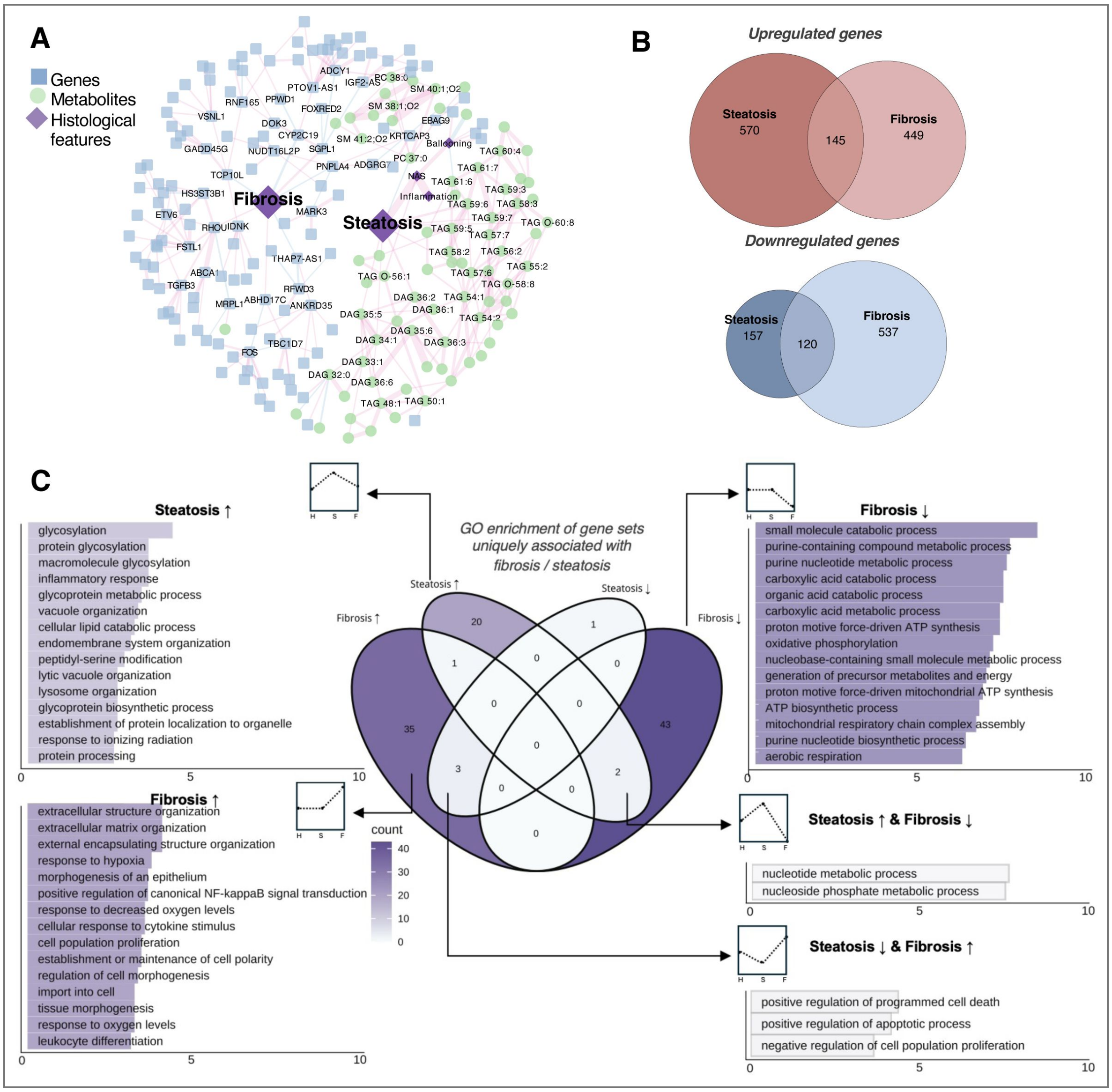


Figure 5. **Steatosis and fibrosis as independent processes.**

(A) Network of genes, metabolites, and histological features (fibrosis and steatosis) using partial correlation analysis as described in the methods. (B) Venn diagrams depicting the number of genes significantly associated with steatosis and fibrosis (q value < 0.05). (C) Gene ontology (GO) enrichment of gene sets specific to steatosis or fibrosis.

Upregulated genes appeared to reflect changes from different cell types, whereas downregulated markers were predominantly hepatocyte-specific (Figure 6A, left panel). This suggests that in the early stages of liver fibrosis development, the transcriptional programs are activated in different cell types, while hepatocyte functions potentially become muted during the process. Meanwhile, we observed fibrosis-associated upregulation of genes involved in TGF-β and SMAD signaling cascades (*TGFB1*, *TGFB3*, and *INHBA*), ECM activators [64–66] (*ITGAV*, *LOXL2*, *THBS1*, *MMP9*, and *MMP14*), regulators (*CDK8*, *CTDSP2*, *HDAC1*, and *HDAC7*), and downstream targets (*TIMP1*, *MMP9*, and *COL5A2*) (Figure S6B). This aligns well with previous reports that TGF-β signaling and HSC activation drive fibrosis during MASLD progression [67, 68].

Given that the transition from simple steatosis to the onset of fibrosis marks a critical window for disease management, we compared gene expression profiles between individuals with fibrosis accompanied by steatosis and those with steatosis but without fibrosis (Table S12). Notably, the top-enriched pathways in this comparison emerged as GTPase signaling and its regulation, signal transduction, and innate immune response (Figure 6B). Moreover, we identified 37 GTPase-related genes displaying significant associations with fibrosis progression (Figure S7A and Table S13). To explore the potential role of GTPase signaling in fibrosis, we examined an inventory of 251 genes encoding GTPases, GTPase-activating proteins (GAPs), and guanine nucleotide exchange factors (GEFs), analyzing their interactome using the Human Cell Map [69] and a protein-protein interaction network[70, 71].

Among the 6,971 proteins that have been tested for physical interaction or subcellular proximity with GTPase-related genes, 508 of these associated significantly with fibrosis progression in our cohort (Figure 6C). The network indicates that GTPases and their regulators interact with a wide range of fibrosis genes, with *ARF3* and *RAC1* being central hubs of the network. Functional enrichment of the core nodes revealed that the GTPase network regulates intracellular transport, actin cytoskeleton organization, exocytosis, and other biological processes during fibrosis progression (Figure S7B). Therefore, GTPases and their regulators are co-regulated with fibrosis-related genes encoding their protein interaction partners, supporting the likelihood of a functional link between GTPases and hepatic fibrosis. This is supported by TGF-β genes positively correlated with co-expressed GTPase-related genes (Figure S7C), suggesting a potential link between TGF-β signaling and GTPase pathways.

To experimentally validate our findings, the altered expression of GTPase-related genes was explored in an independent system using an established model of 3D liver spheroids in which hepatic cells remain viable and functional for multiple weeks [72–74]. Specifically, we co-cultured primary human hepatocytes with Kupffer and stellate cells isolated from patients with histologically confirmed MASH [75] and examined the overall expression profile of GTPase-regulated genes between our human cohort and the ex vivo system. The expression of GTPases were largely concordant when assessing human patients and the 3D spheroid system (Figure 6D). In comparison to the control group, spheroids from MASH patients demonstrated an upregulation of 24 genes encoding GTPases and their regulators (Figure 6E). Remarkably, the treatment of liver spheroids with elafibranor, a dual PPARα/δ agonist [76], restored the expression levels of 17 (70%) GTPase-related genes back to the baseline (Figure 6E). This independent system further suggests that GTPases may play a role in MASLD pathogenesis and the fibrotic response.

Since HSCs are the main liver cell type responsible for activating fibrosis, we next assessed whether inhibition of GTPase activity in an immortalized HSC cell line, the LX-2 cells, could attenuate markers of fibrogenesis (Figure S8A). Selective GTPase inhibitors targeting Rac1 (NSC23766) and Cdc42 (ML141) reduced the mRNA expression of *COL1A1* and *COL1A2*, as well as pro-collagen secretion (Figure S8) under basal conditions. TGF-β1 mediated activation induced the expected increase in collagen secretion and gene expression which was attenuated by Rac1 inhibitor (NSC23766) and to a lesser extent by ML141. In addition, examination of previously published transcriptomic data of HSCs isolated from CCL_4-mediated liver fibrosis in mice [77], revealed the upregulation of GTPases comparable to the steatosis-to-fibrosis transition in our human cohort, with a temporal pattern aligning with hepatic collagen deposition (Figure S9A).

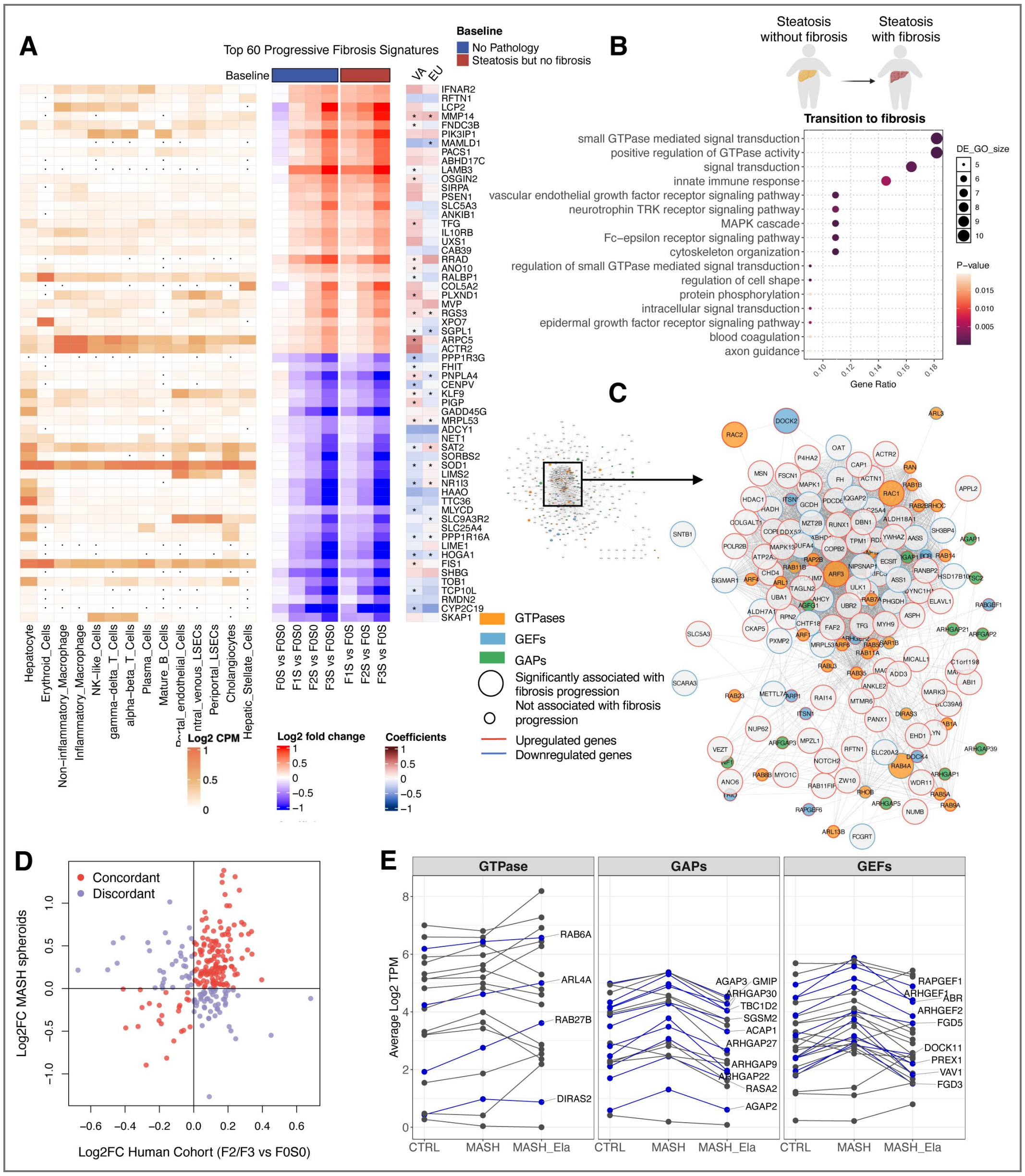


Figure 6. Liver fibrosis signatures and potential therapeutic targets of fibrosis initiation.

(A) Top 60 fibrosis markers in the liver. A total of 213 genes were identified, from which the top 30 upregulated and top 30 downregulated genes are visualized. The middle plot (blue bar on top) shows the comparison of individuals with fibrosis and without pathology (baseline - “no MASLD”). For the right plot (red bar), individuals with fibrosis were compared to those with steatosis but no fibrosis (baseline – “steatosis but no fibrosis”). Results were cross-referenced with two published cohorts (VA cohort, GSE130970; EU cohort, GSE135251, right). Asterisks indicate genes with a q-value < 0.05 and a consistent change in direction within our cohort. The average gene expressions of liver cell types were obtained from GSE115469 at the log2CPM level (left). Dots in the single cell map indicate zero expression in the corresponding cell types. (B) Enrichment of pathways in the transition from simple steatosis to the onset of fibrosis. (C) Protein-protein interaction network of GTPases and their regulators. Nodes are colored based on gene type, with borders indicating the direction of gene regulation. Node size corresponds to the significance of the genes in relation to fibrosis grades within this cohort. GEFs: Guanine nucleotide exchange factors. GAPs: GTPase-activating proteins. (D) Comparison of expression level changes in GTPase-related genes between this human cohort and an independent 3D spheroid MASH system. Log2 fold change for the human cohort was calculated by comparing patients with grade 2 or 3 fibrosis to those without fibrosis or steatosis. Genes with the same direction of change are colored in red, while others are colored in purple. (E) Expression of GTPase-related genes in patient-derived 3D liver spheroids: control spheroids, spheroids from patients with MASH, and Elafibranor-treated MASH spheroids (n = 4). Sixty-eight genes with a p-value < 0.05 from the ANOVA test were plotted in the diagram, with blue lines highlighting 24 genes that exhibited increased expression in the MASH group compared to the control group.

Further, in a human liver organoid model, TGF-β induced increased expression of GTPase-related genes in hepatocytes and HSCs, but not in fibroblasts [78] (Figure S9B), suggesting a potential feedforward loop involving the TGF-β/GTPase axis between hepatocytes and HSCs. To investigate intercellular crosstalk in GTPase regulation, we examined key GTPase-related genes in LX-2 cells, hepatocyte monocultures, and spheroid co-cultures (including hepatocytes, HSCs, Kupffer cells). As shown in Figure S10, TGF-β1 induced a potential increase in *VAV1* and *DOCK2* consistent in co-cultures, hepatocytes, and LX-2 cells, while *RAC1*, *RAB32*, and *RHOU* showed cell type-specific responses. These findings indicate that multiple hepatic cell types mediate GTPase regulation, underscoring intercellular crosstalk.

Overall, these additional experiments and analyses of datasets identified upregulated GTPase-related genes during fibrosis initiation (Figure S9C). However, further in-depth mechanistic studies are needed to validate this association.

Discussion

In this study, we performed a comprehensive omics-based analysis of a cohort of 109 obese individuals with early MASLD. Through integrative dual-omics approaches, we mapped the liver and plasma metabolomes and identified distinct hepatic molecular features associated with steatosis and fibrosis progression. Notably, fibrosis was closely connected to global reprogramming of hepatic gene expression, with GTPase-related genes emerging as possible mediators of fibrosis initiation.

In this obesity cohort, the plasma metabolome showed limited alterations in MASLD, with the exception of elevated circulating TAGs. This contradicts with McGlinchey *et al.*, who reported more than 20 differential metabolites in the serum metabolome [19]. Our conjecture is that the discrepancy stems from the different disease spectrum covered in the two studies, as the cohort in the other study included advanced MASLD cases with NAS score ≥ 4 (63.4%). In addition, our study focused exclusively on obese individuals (median BMI > 43.0), substantially higher than the other study. A recent report has highlighted distinct metabolite signatures between obese and lean individuals with MASLD, noting fewer differential metabolites in obese individuals than lean ones [79]. This suggests that plasma lipidome is less reflective of liver pathology in patients with obesity compared to lean individuals. Similarly, a BMI-stratified multivariable model revealed distinct circulating MASLD metabolite signatures across different degrees of obesity, with lean subgroups showing significantly altered circulating lipotoxic intermediates [80]. Therefore, as shown in Figure 2C, we hypothesize that co-occurring morbidities in obesity may obscure liver-derived metabolic signals in the plasma, particularly in early-stage MASLD. Given that obese individuals represent the predominant population affected by MASLD, it is reasonable to conclude that whether blood analysis of metabolites attains diagnostic accuracy in identifying disease progression at early stages depends on the disease spectrum covered in a study and remains context-specific. However, the emergence of liquid biopsies, fine needle liver biopsies, and the evaluation of liver-specific extracellular vesicles may open up other minimally invasive diagnostic opportunities [81].

The strength of our study is the characterization of hepatic pathophysiology in relation to steatosis and fibrosis via transcriptome and metabolome-wide variations. Consistent with general expectation, our work highlighted altered hepatic lipid metabolism in human liver tissues of obese patients at both metabolite and gene levels [82–84]. Consistent with the variations in GLs, higher expression levels of *DGAT2*, *PNPLA3*, and *PLIN3* were associated with steatosis progression. The higher expression of *DGAT2*, which encodes a key enzyme catalyzing the last step of de novo TAG synthesis, implies enhanced integration of TAGs into LDs in the endoplasmic reticulum, which presumably alleviates lipid-induced ER stress and evades accumulation of lipotoxic lipids [85]. Upregulation of *PLIN3* was correlated with higher grades of both fibrosis and steatosis. The protein encoded by *PLIN3*, along with other PLIN proteins [86], plays an important role in LD stabilization and the prevention of TAG hydrolysis, thus potentially contributing to hepatic steatosis [87]. In addition, although previous animal studies have reported increased levels of sphingolipids in MASLD models [19, 88], we did not observe this in patients with obesity at early stages of the disease. Sphingolipid alterations varied across cohorts with differing patient compositions [18, 19, 89], suggesting that sphingolipid metabolism may be dependent on disease stage, obesity status, and exhibit discrepancies between humans and mice.

Beyond lipid metabolism, other metabolic pathways and processes also exhibited dysregulation at the dual-omics level. Specifically, with excessive lipid deposition, we observed higher levels of taurine and vitamin E, both of which have antioxidant effects and likely reflect the hepatic response to oxidative stress [90–92]. Importantly, amino acid metabolism was markedly dysregulated as steatosis progressed, characterized by elevated hepatic levels of serine, arginine, glutamic acid, glycine, and alanine, along with decreased levels of the aromatic amino acids phenylalanine, tryptophan, and tyrosine. A previous *in vitro* study also demonstrated that palmitic acid supplementation disrupted the metabolism of the aromatic amino acids in PH5CH8 and HepG2 cells [93]. As microbiota-derived metabolites, dysregulation in these amino acid may reflect disrupted gut function and may affect the de novo synthesis of nicotinamide adenine dinucleotide in MASLD patients [94]. Moreover, we also observed evidence of autophagy activation concurrent with intrahepatic lipid and cholesterol accumulation, wherein lysosome-associated genes were uniquely associated with steatosis, as shown in Figure 4B and highlighted in Figure 5C by the enrichment of lysosome organization among steatosis-specific processes. This autophagic response most likely represents a protective mechanism to ameliorate steatosis [95]; however, the downregulation of autophagy regulators during fibrosis development may exacerbate liver metabolic dysfunction.

As liver fibrosis progressed, the expression levels of genes involved in primary bile acid biosynthesis, PUFA metabolism, and steroid hormone biosynthesis were lower, especially *ACADS*, *ACADSB*, and *ACADM* which encode key enzymes in the initial stage of catalyzing fatty acid β-oxidation, and *HADH* and *ACAA1*, which encode enzymes that act in the late stage of β-oxidation [96]. Moreover, mitochondrial-related genes were downregulated with fibrosis progression, particularly those related to the electron transport chain of the mitochondrial inner membrane. This may indicate that pathological oxidative stress and impaired ATP production caused by early mitochondrial dysfunction may aggravate fatty liver inflammation [14], and the augmentation of the inflammatory response may become an important cause of liver fibrosis.

Through univariate hypothesis testing and partial correlation analysis, we revealed distinct molecular signatures correlated with fibrosis and steatosis. Liver fibrosis was associated with gene expression alteration, from which we identified over 200 progressive markers tracking its gradual advancement. Furthermore, GTPase-related genes were dysregulated at the mRNA level with the emergence and progression of fibrosis. Increased expression of GTPase regulators was independently confirmed in 3D primary human liver spheroids, with elafibranor restoring their expression to baseline levels (Figure 6E). The downregulated expression of GTPase regulators following elafibranor treatment suggests a potential anti-fibrotic mechanism involving GTPase signaling.

GTPase proteins are categorized into several subfamilies such as Rho-, Ras-, and Arf-GTPase based on their sequence and structure [97]. GAPs and GEFs are the key regulators of GTPase activity that modulate intrinsic GTPase functions and the GTP-bound state [98]. Only a few studies have elucidated the role of GTPases in liver pathology [99–102]. For instance, *Rap1a* was identified as a signaling molecule that suppresses both gluconeogenesis and hepatic steatosis [102]. The Rab23-specific GAP, *GP73*, has been implicated in triggering non-obese MASLD [101]. Additionally, Rho-GTPase signaling through the ROCK1/AMPK axis regulates de novo lipogenesis during overnutrition [99], while an immune-related GTPase triggers lipophagy and prevents hepatic lipid storage [100]. However, evidence regarding the role of GTPase regulation in human liver is limited, particularly in the context of liver fibrosis initiation and progression, and a thorough mechanistic investigation is warranted to establish causality.

In recent years, dual- and multi-omics strategies have gradually emerged in MASLD research. A proteo-transcriptomic map constructed from the EU cohort's liver RNA sequencing data and circulating proteome identified four promising protein biomarkers [28]. Among these, *AKR1B10* was also identified in our study as a key gene related to lipid metabolism, showing strong associations with both steatosis and fibrosis (Figure 3A), which highlights its potential as a biomarker even in the early stages of MASLD. Additionally, we observed mitochondrial dysregulation linked to the progression of liver fibrosis (Figure 4A). Consistent with our findings, a multi-omics study in obese individuals with MASLD also revealed mitochondrial dysfunction at the hepatic protein level. The study further emphasized similar mitochondrial disruptions in adipose tissue, suggesting the liver-adipose tissue interaction in obesity-related MASLD [103]. Beyond inter-tissue interactions, Raverdy *et al*. reported the heterogeneity of MASLD molecular signatures in relation to varying degrees of cardiovascular risk and type 2 diabetes [104]. This further highlights the impact of systemic metabolic conditions on MASLD progression. In our analysis of diabetic individuals, we identified more downregulated metabolic genes associated with fibrosis, supporting the notion that insulin resistance may exacerbate metabolic dysregulation and interferes with the disease trajectory.

Overall, our integrative investigation offers one of the most detailed molecular landscapes of early stage MASLD in individuals with obesity, with comprehensive descriptors of the early remodeling of hepatic metabolism associated with initiation of fibrosis in the steatotic liver.

Limitations of the study

This study focused on 109 patients with early-stage MASLD, potentially overlooking molecular changes associated with later disease stages. To address this, we cross-referenced our findings with two external cohorts (EU and VA). Additionally, we identified GTPase-related genes as potential future targets for reversing liver fibrosis. While preliminary validation has been conducted in different systems, further investigation is required to confirm the causal role of specific GTPases in fibrosis initiation especially in HSCs and to elucidate how GTPase signaling contributes to collagen production throughout the progression of MASLD.

Materials and Methods

Ethics statement

Participants provided written and verbal informed consent. The study protocol conforms to the ethical guidelines of the 1975 Declaration of Helsinki and was approved by the University of Melbourne Human Ethics Committee (ethics ID 1851533), The Avenue Hospital Human Research Ethics Committee (Ramsay Health; ethics ID WD00006, HREC reference number 249), the Alfred Hospital Human Research Ethics Committee (ethics ID GO00005), and Cabrini Hospital Human Research Ethics Committees (ethics ID 09-31-08-15). All work on the data from human samples in Sweden was approved by Etikprövningsmyndigheten (Dnr 2024-07917-01).

Cohort recruitment

A detailed medical history was taken, and metabolic comorbidities were noted including the presence of previously diagnosed hypertension and diabetes assessed by oral glucose tolerance testing. Exclusion criteria included: age <18 years, previous gender reassignment, other causes of chronic liver disease and/or hepatic steatosis including Wilson's disease, α-1-antitrypsin deficiency, viral hepatitis, human immunodeficiency virus, primary biliary cholangitis, autoimmune hepatitis, genetic iron overload, hypo- or hyperthyroidism, coeliac disease, as well as recent (within three months of screening visit) or concomitant use of agents known to cause hepatic steatosis including corticosteroids, amiodarone, methotrexate, tamoxifen, valproic acid and/or high dose oestrogens [105]. Further exclusion criteria included potential for alcohol-induced liver disease, which was assessed through a modified version of the alcohol use disorders identification test (AUDIT) [105, 106]. Following histological assessment, patients with liver slice weighing less than 2mg were excluded from the omics analysis.

Eligible patients with obesity scheduled for primary or secondary sleeve gastrectomy, gastric bypass, or the insertion of a laparoscopic-adjustable band were prospectively enrolled. All patients were fasted for 8-12 h overnight and venous blood was taken before the induction of anaesthesia. Blood was transferred to 2 x K2E Ethylenediaminetetraacetic acid (EDTA), 2 x SST™ II Advance and

1 x FX 5 mg bio-containers for subsequent storage or clinical/biochemical assessments. All blood samples were sent to Melbourne PathologyTM (Victoria, Australia) for standardised measurement of biochemical and metabolic variables, except for one bio-container of EDTA. Standard blood analyses were performed for electrolytes, full blood examination, glucose, glycosylated haemoglobin, insulin, C-peptide, cholesterol, triglycerides, and liver function assessed by ALT, AST, GGT, and alkaline phosphatase (ALP), and screening blood tests for liver disease. The remaining blood within the EDTA tube was spun at 8000 x g for 10 min and the plasma was collected and stored at -80°C for mass spectrometry analyses.

Liver biopsy collection and histological feature assessment

An ~ 1 cm^3 wedge liver biopsy was collected from the left lobe of the liver during surgery. The liver was cut into two portions. One portion was placed in formalin and transported to TissuPath™ (Mount Waverley, Victoria), paraffin embedded and processed for histological analysis. Samples were graded according to the Clinical Research Network NAFLD activity score (NAS) [107] and Kleiner classification of liver fibrosis [108] by a liver pathologist at TissuPath. Patients with no MASLD were defined by steatosis score of 0, regardless of inflammation or fibrosis grade 1. MASL patients were defined by a steatosis score ≥1 with or without lobular inflammation. MASH was defined by the joint presence of steatosis, lobular inflammation, and ballooning[108]. The remaining portion of liver slices was used for bulk RNA sequencing and untargeted metabolomics analysis.

Sample preparation for untargeted metabolomics

For liver tissue biopsies, each sample containing 50 ± 5 mg of tissue was extracted in 0.8 mL 50:50 (v/v) methanol:chloroform [109, 110]. Samples were homogenized for 10 min at 25 Hz with a single 3-mm tungsten carbide bead per tube. Separation of phases was achieved by the addition of 0.4 mL of water followed by vortex-mixing and centrifugation (2400 g, 15 min, 4 °C). After separation, the upper phase (the metabolite-containing fraction) and the lower phase (the lipid-containing fraction) were transferred into separate tubes and dried using SpeedVac.

For plasma samples, 225 µL of methanol was added to 50 µl plasma, and the mixture was vortexed for 10 sec [111]. Subsequently, 450 µL chloroform was added and the mixture was incubated for 1h in a shaker. To induce phase separation, 187.5 µL water was added and the mixture was centrifuged at 12,000 rcf for 15 min at 4 °C. The upper phase (the metabolite-containing fraction) phase and lower (the lipid-containing fraction) were collected into separate fresh tubes and dried using SpeedVac.

Dry extracts from the organic fraction were resuspended in the mixed solvent of 90:10 isopropanol: acetonitrile. For the aqueous fraction, 90% acetonitrile was used to reconstitute dry extracts. Plasma samples were reconstituted in 1:4 dilution (plasma-to-solvent: 50 µL/200 µL), and the reconstitution volumes for liver biopsy samples were corrected by the weights of liver slices (tissue-to-solvent: 50 mg/1000 µL). After reconstitution, samples were sonicated for 15 min and centrifuged at 12,000 g for 10 min at 4 °C.

Metabolomics data acquisition and processing

A 1 µL aliquot of the extract was subjected to LC-MS analysis using the Sciex TripleTOF 6600 system coupled with the Agilent 1290 HPLC. For polar metabolite analysis, mobile phase A was water with 10 mM ammonium formate and mobile phase B was acetonitrile with 0.1% formic acid. Flow rate was 0.250 ml/min. The LC gradient started at 3% A, increased to 30% A from 1 to 12 min, then to 90% A from 12 to 15 min, remained at 90% until 18.5 min, and returned to 3%, holding until the end of the 24-min run. For lipid analysis, mobile phase A was prepared by 60:40 water: acetonitrile with 10 mM ammonium formate, and mobile phase B was made by 90:10 isopropanol: acetonitrile with 10 mM ammonium formate. Flow rate was 0.250 ml/min and the total run time is 15.80 min. The LC gradient started at 80% A, decreased to 40% A at 2 min, further decreased to 0% from 2 to 12 min, remained at 0% A until 14 min, then returned to 80% A at 14.10 min, and held at 80% A until the end of run. Mass spectrometry settings were as follows: gas1 50 V, gas2 60V, curtain gas 25V, source temperature 500 °C, IonSpray voltage 5500 V. The collision energy was set to 30 V and 45 V for polar metabolites and lipids, respectively. Fifteen precursors were selected for MSMS fragmentation in the DDA acquisition. The mass range for the RP method was 300 – 1000 m/z and for the HILIC method was 50 – 1000 m/z. In the SWATH acquisition, fixed windows were applied with window widths of 21 Da for the HILIC method and 10 Da for the RP method.

Each tissue type (i.e., plasma and liver) was analyzed as a separate batch. Following sample reconstitution, the supernatants were collected into MS vials and pooled as quality control (QC) samples for each tissue type. For each analytical batch, patient samples were injected in a randomized sequence, along with one QC injection approximately every 10 samples for monitoring instrument stability and calculating the coefficient of variation (CoV). Both reverse phase (RP) and hydrophilic interaction liquid chromatography (HILIC) columns were used for chromatographic separations of organic and aqueous fractions, respectively. For the analysis of aqueous fraction, metabolites were separated on SeQuant ZIC-cHILIC (3 μm, 100Å, 100x2.1 PEEK) column with a flow rate of 0.25 mL/min in a 24-minute run. For the organic fraction, metabolites were separated on RRHD Eclipse Plus C18 (2.1 x 50 mm, 1.8 μm).

Data acquisition was performed in both positive and negative ionization modes. Data-dependent acquisition (DDA) was performed on pooled QC samples for compound identification and SWATH-MS analysis was applied to individual samples for relative quantification. The mass range for the RP method was 300 – 1000 m/z and for the HILIC method was 50 – 1000 m/z. Data was converted into mzML format and was then processed by MetaboKit [112] (*https://github.com/MetaboKit*). To gain broad metabolite coverage, we curated a data processing pipeline for metabolites identified via spectral matches (MSMS-level, with 915 identifications postprocessing) and those identified solely through mass matches (MS1-level, with 548 identifications postprocessing). For metabolite quantification, the project-specific spectral library generated from DDA analysis was used to extract MS1 and MS2 features from the SWATH data using the MetaboKit DIA module. Compound transitions with a CoV > 30% were excluded from the analysis. For each compound, the precursor or fragment ion with the lowest CoV was selected as the quantifier. For a compound identified in both positive and negative modes using the same column, the peak feature with the lower CoV were selected. Overlapping metabolites in the RP and HILIC datasets are mainly semi-polar lipids, as RP detects compounds with m/z > 300. For lipids identified by both methods, the ones measured on the RP column was selected.

RNA sequencing

Transcriptome sequencing service was provided by Biomarker Technologies (BMK), GmbH. Briefly, RNA were extracted from liver biopsies using Trizol reagent. The quality and quantity of extracted RNA was assessed by Nanodrop (quantity and purity) and Labchip GX (Quality). The cDNA libraries were prepared using Hieff NGS Ultima Dual-mode mRNA Library Prep Kit from Illumina (Yeasen) as per manufacturer's instructions. RNA sequencing was performed on NoveSeq6000 platform (PE 150 mode). For raw data processing, FastQC (v0.11.9) was used as a quality control check for raw sequencing. Alignment to the reference genome (GRCh38, Ensembl release 77) was performed using STAR (v2.7.9a), and gene expression levels (transcripts-per-million, TPM) were estimated using RSEM (v1.3.1). Gene expression data underwent a logarithmic transformation to log2(TPM + 1) and mean-centering normalization for downstream analyses.

Statistical analysis

All analyses in this study were conducted using R version 4.3.1. Linear regression model was used to evaluate the linear association between histological grades and molecular abundance levels with adjustment for age, sex, BMI, and diabetes. Statistically significantly changed metabolites or genes were defined by controlling the overall type I error at 5% or 10% (q value < 0.05 or 0.1). Spearman rank correlation was used to assess the relationship between the same metabolite in the liver and plasma. This accounts for matrix effects across different tissue types in untargeted analysis by comparing the relative intensity ranking of a metabolite across all samples within each tissue type. Gene overrepresentation analyses were performed using gene ontology recourse by in-house software [113]. ClueGO, a plugin app in Cytoscape, was used to visualize enrichment maps [114]. Partial correlation network analysis was employed to integrate the two omics layers using an in-house partial correlation network algorithm ACCORD [31]. KEGG pathway database was used for the analysis of lipid metabolism and pathway mapping [115]. All network visualizations in this study were done using Cytoscape [116].

To investigate MASLD subgroup signatures related to diabetic status, we analyzed linear associations between gene signatures and histological features separately in non-diabetic (n = 71) and diabetic individuals (n = 23). Statistical power was estimated by comparing variance explained in full (y ~ x + a + b + c) versus reduced (y ~ a + b + c) regression models, converting the incremental R^2 into Cohen's f^2, and applying pwr.f2.test at $\alpha = 0.05$. We further compared gene expression profiles between diabetic (n = 21) and non-diabetic (n = 43) MASLD patients, identifying 166 differentially expressed genes ($p < 0.05$, $|\log_2 FC| > 0.32$). Of these, 54 genes (Table S10) were both differentially expressed and significantly associated with fibrosis progression, and thus marked as signatures in diabetic MASLD.

To identify progressive markers of liver fibrosis, we performed pairwise analyses for fibrosis stages using two distinct baselines: patients with no MASLD and those with steatosis but no fibrosis. 213 markers were characterized as progressive based on the following criteria: a) significant linear association with fibrosis grades, and b) significant differential expression (p value < 0.05 in t-test) in at least two fibrosis stages (e.g., F1 and F2, F2 and F3) compared to both baselines.

To identify independent gene signatures associated specifically with steatosis and fibrosis progression, we applied linear regression models for each outcome adjusting for the other. Steatosis-specific signatures were defined as genes significantly associated with steatosis but not with fibrosis, and vice versa. Metascape was used for meta-analysis of enrichment pathways [117]. Pathways with at least 10 hits and p value < 0.05 were considered steatosis- or fibrosis-specific.

For meta-analysis of reference human cohorts, the two RNA-seq datasets of the liver transcriptome, the VA cohort (GSE130970) and the EU cohort (GSE135251), were downloaded from Gene Expression Omnibus, and associations between gene expression (log2 TPM) and fibrosis grades or NAS were analyzed using linear regression. In the EU cohort, the patients without MASH diagnosis were removed from the meta-analysis for the purpose of identifying fibrosis signatures in MASH patients. Human liver single cell dataset was obtained from GSE115469 [63]. The mean log2 TPM was calculated for each cell type, and cell specificity was determined by subtracting the mean expression levels across all cell types.

Cross-referencing datasets of mouse models and human liver organoids

The RNA sequencing data (raw counts) for HSCs isolated from liver fibrosis mouse models [77] was downloaded from GSE176042 and analyzed using the DESeq2 R package. This previously published study investigated HSC initiation and perpetuation using acute and chronic models, respectively. For the acute model, mice were injected with CCl_4 once and samples were taken at 24 h, 72 h, and 1 w. For the chronic model, mice received a regimen of semi-weekly injection of CCl_4 for 4 weeks. After this regimen, mice were sacrificed and samples were taken at 24 h, 72 h, and 2 w. The statistical results from single cell dataset of stem cell-derived human liver organoids were obtained from GSE207889 [78].

3D liver spheroid MASH model

3D liver spheroids were generated based on co-cultures of cryopreserved primary human hepatocytes (PHH) and liver non-parenchymal cells (NPC) as previously described [75]. Briefly, cells were seeded at a PHH:NPC ratio of 6:1 in 96-well ultra-low attachment plates (CLS7007-24EA Corning) with a total of 1500 cells/well in culture medium (William's E medium containing 11 mM glucose, 100 nM dexamethasone, 10 µg/mL insulin, 5.5 mg/L transferrin, 6.7 µg/L selenite, 2 mM L-glutamine, 100 U/mL penicillin, 0.1 mg/mL streptomycin) supplemented with 10% FBS. After spheroid formation FBS was phased out and cells were maintained in medium supplemented with albumin-conjugated free-fatty acids (FFA) including 80μM palmitic acid and 80 μM oleic acid. At the end of the experiment, spheroids were collected for RNA extraction (Zymo R105). The elafibranor treatment protocol was described in the previous study [75]. Each experimental condition (control, MASH, and elafibranor treatment) was performed in quadruplicate.

Data availability

All data are available within the manuscript, supplemental figures, and supplemental tables. RNA sequencing data to this article have been submitted to SRA (PRJNA1185558) and is deposited with GEO (GSE281797). Untargeted LC-HRMS data is deposited with Zenodo, including IDA data (DOI 10.5281/zenodo.14091962), SWATH data for the organic fraction (DOI 10.5281/zenodo.14096635), and SWATH data for the aqueous fraction (DOI 10.5281/zenodo.14096753, DOI 10.5281/zenodo.14136832). All clinical data, processed omics datasets, and code are available at *https://github.com/SLINGhub/MASLD_dual_omics*.

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Additional information

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Author contributions

QZ*: designing research studies, conducting experiments, acquiring data, analyzing data, writing the manuscript

WD*: designing research studies, conducting experiments, acquiring data, analyzing data, writing the manuscript

RW*: analyzing data, writing the manuscript

YZ: conducting experiments, acquiring data, analyzing data UK: analyzing data, writing the manuscript

LNZ: analyzing data, writing the manuscript

HT: conducting experiments, acquiring data, analyzing data

SY: conducting experiments, acquiring data, analyzing data

MY: conducting experiments, acquiring data, analyzing data

YX: conducting experiments, acquiring data, analyzing data

YK: conducting experiments, acquiring data, analyzing data

SL: conducting experiments, acquiring data, analyzing data

RL: conducting experiments, acquiring data, analyzing data

GT: conducting experiments, acquiring data, analyzing data

PN: conducting experiments, acquiring data, analyzing data

PRB: providing reagents, cohort management

VMK: designing research studies, analyzing data, writing the manuscript, supervision, funding acquisition

HC: designing research studies, analyzing data, writing the manuscript, supervision, funding acquisition

MJW: designing research studies, analyzing data, writing the manuscript, supervision, funding acquisition

PK: designing research studies, analyzing data, writing the manuscript, supervision, funding acquisition, leading of collaboration

QZ, WD, and RW are co-first authors based on their contributions, which was writing original draft, LC-MS data acquisition and bioinformatic analysis of transcriptomic, metabolomic, and clinical data (QZ), organizing cohort samples, clinical parameters, managing, and LX-2 experiments (WD), and writing the original draft and analyzing data (RW).

ALT: Alanine aminotransferase
AST: Aspartate aminotransferase
ALP: Alkaline phosphatase
CAR: Acylcarnitines
CCl_4: Carbon tetrachloride
Cer: Ceramide
CM: Conditioned medium
CYP: Cytochrome P
DAG/DG: Diacylglycerol
DDA: Data-dependent acquisition
ECM: Extracellular matrix
FBS: Fetal bovine serum
FFA: Free-fatty acids
FXR: Farnesoid X receptor
GAP: GTPase-activating protein
GEF: Guanine nucleotide exchange factor
GGT: Gamma-glutamyltransferase
GL: Glycerolipid
GPL: Glycerophospholipid
HSC: Hepatic stellate cell
HILIC: Hydrophilic interaction liquid chromatography
LD: Lipid droplet
LNAPE: N-acyl-lysophosphatidylethanolamine
LPC-O: Ether-linked lysophosphatidylcholine
LPI: Ether-linked lysophosphatidylinositol
LPS: Ether-linked lysophosphatidylserine

LPC: Lysophosphatidylcholine
LPE: Lysophosphatidylethanolamine
MG: Monoacylglycerol
MASH: Metabolic dysfunction-associated steatohepatitis
MASLD: Metabolic dysfunction-associated steatotic liver disease
NAS: NAFLD activity score
NPC: Non-parenchymal cells
NAE: N-acyl ethanolamines
PCA: Principal component analysis
PHH: Primary human hepatocytes
PUFA: Polyunsaturated fatty acid
PC: Phosphatidylcholine
PC-O: Ether-linked phosphatidylcholine
PE: Phosphatidylethanolamine
PE-O: Ether-linked phosphatidylethanolamine
PG: Phosphatidylglycerol
PI: Phosphatidylinositol
PS: Phosphatidylserine
QC: Quality control
RP: Reverse phase
SM: Sphingomyelin
SL: Sphingolipid
SPB: Sphingoid base
TAG/TG: Triacylglyceride
TAG-O: Ether-linked triacylglycerol
Vitamin E: Alpha-tocopherol

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Additional files

Combined Supplementary text and data

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Peer reviews

Reviewer #1 (Public review):

Summary:

Metabolic dysfunction-associated steatotic liver disease (MASLD) ranges from simple steatosis, steatohepatitis, fibrosis/cirrhosis, and hepatocellular carcinoma. In the current study, the authors aimed to determine the early molecular signatures differentiating patients with MASLD associated fibrosis from those patients with early MASLD but no symptoms. The authors recruited 109 obese individuals before bariatric surgery. They separated the cohorts as no MASLD (without histological abnormalities) and MASLD. The liver samples were then subjected to transcriptomic and metabolomic analysis. The serum samples were subjected to metabolomic analysis. The authors identified dysregulated lipid metabolism, including glyceride lipids, in the liver samples of MASLD patients compared to the no MASLD ones. Circulating metabolomic changes in lipid profiles slightly correlated with MASLD, possibly due to the no MASLD samples derived from obese patients. Several genes involved in lipid droplet formation were also found elevated in MASLD patients. Besides, elevated levels of amino acids, which are possibly related to collagen synthesis, were observed in MASLD patients. Several antioxidant metabolites were increased in MASLD patients. Furthermore, dysregulated genes involved in mitochondrial function and autophagy were identified in MASLD patients, likely linking oxidative stress to MASLD progression. The authors then determined the representative gene signatures in the development of fibrosis by comparing this cohort with the other two published cohorts. Top enriched pathways in fibrotic patients included GTPas signaling and innate immune responses, suggesting the involvement of GTPas in MASLD progression to fibrosis. The authors then challenged human patient derived 3D spheroid system with a dual PPARa/d agonist and found that this treatment restored the expression levels of GTPase-related genes in MASLD 3D spheroids. In conclusion, the authors suggested the involvement of upregulated GTPase-related genes during fibrosis initiation.

Concerns from first round of review:

(1) A recent study, via proteomic and transcriptomic analysis, revealed that four proteins (ADAMTSL2, AKR1B10, CFHR4 and TREM2) could be used to identify MASLD patients at risk of steatohepatitis (PMID: 37037945). It is not clear why the authors did not include this study in their comparison.

(2) The authors recruited 109 patients but only performed transcriptomic and metabolomic analysis in 94 liver samples. Why did the authors exclude other samples?

(3) The authors mentioned clinical data in Table 1 but did not present the table in this manuscript.

(4) The generated metabolomic data could be a very useful resource to the MASLD community. However, it is very confusing how the data was generated in those supplemental tables. There is no clear labeling of human clinical information in those tables. Also, what do those values mean in columns 47-154? This reviewer assumed that they are the raw data of metabolomic analysis in plasma samples. However, without clear clinical information in these patients, it is impossible that any scientist can use the data to reproduce the authors' findings.

(5) In Fig. 5B, the authors excluded the steatosis and fibrosis overlapped genes. Steatosis and fibrosis specific genes could simply reflect the outcomes rather than causes. In this case, the obtained results might not identify the gene signatures related to fibrosis initiation.

(6 In Fig. 6D, the authors used 3D liver spheroid to validate their findings. However, there is no images showing the 3D liver spheroid formation before and after PPARa/d agonist treatment. It is not clear whether the 3D liver spheroid was successfully established.

(7) The authors suggested that targeting LX-2 cells with Rac1 and Cdc42 inhibitors could reduce collagen production. Did the authors observe these two genes upregulated in mRNA and protein expression levels in their cohort when compared MASLD patients with and without fibrosis?

(8) Did the authors observe that the expression levels of Rac1 and Cdc42 are correlated with fibrosis progression in MASLD patients?

(9) Other studies have revealed several metabolite changes related to MASLD progression (PMID: 35434590, PMID: 22364559). However, the authors did not discuss the discrepancies between their findings with the previous studies.

Significance:

Overall, the current study might provide some new resources regarding transcriptomic and metabolomic data derived from obese patients with and without MASLD. The MASLD research community will be interested in the resource data.

Comments on revised version:

Thank you for the authors' responses to my concerns. I do not have any further comments.

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Reviewer #2 (Public review):

In this paper, Kaldis and collaborators investigate the molecular heterogeneity of a 109 morbidly obese patient cohort, focusing on liver transcriptomics and metabolomics analysis from liver and serum. The main finding (i.e. upregulation of GTPase-coding genes) was validated in spheroids and a human HSC cell line. As these proteins are involved in critical cellular functions related to metabolism and cytoskeleton dynamics, these findings shed light on their involvement in human liver pathology which so far has been poorly (or even not) documented to date. This is an interesting addition to the current knowledge about chronic liver pathology and warranting further in-depth molecular investigations to address molecular mechanisms of action (cellular specificity, GTPase-driven pathways...).

Strengths:

Using a well-characterized patient cohort of moderate size, the study provide transcriptomic and metabolomic data of high quality with adequate statistical corrections which are a very useful resource for the community. Mechanistic experiments usefully hint at novel druggable targets in the early steps of fibrosis, hence probably in hepatic stellate cell activation.

Weaknesses:

Cross comparisons with other cohorts is informative but of limited interest due to patient classification issues, inherent to histological staging practices. The lack of correlation between transcriptomic and metabolomic data is deceptive but expected due to the systemic nature of metabolomic analysis and was also observed in recently published papers.

Comments on revised version:

I have no further comment about this amended version, aside from suggesting to add (if known) the time at which biopsies were collected. Time-of-day is an important yet often overlooked parameter of gene expression variation, and along the same line, the imposed fasting to bariatric surgery patients is also a matter of variation of gene expression and of metabolite abundance. It is hoped that future investigations will more precisely characterize the role of the newly identified targets in MASLD.

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Reviewer #3 (Public review):

Summary:

Metabolic dysfunction associated liver disease (MASLD) describes a spectrum of progressive liver pathologies linked to life style-associated metabolic alterations (such as increased body weight and elevated blood sugar levels), reaching from steatosis over steatohepatitis to fibrosis and finally end stage complications, such as liver failure and hepatocellular carcinoma. Treatment options for MASLD include diet adjustments, weight loss, and the receptor-β (THR-β) agonist resmetirom, but remain limited at this stage, motivating further studies to elucidate molecular disease mechanisms to identify novel therapeutic targets.

In their present study, the authors aim to identify early molecular changes in MASLD linked to obesity. To this end, they study a cohort of 109 obese individuals with no or early-stage MASLD combining measurements from two anatomic sides: 1. bulk RNA-sequencing and metabolomics of liver biopsies, and 2. metabolomics from patient blood. Their major finding is that GTPase-related genes are transcriptionally altered in livers of individuals with steatosis with fibrosis compared to steatosis without fibrosis.

Comments from the first round of review:

(1) Confounders (such as (pre-)diabetes)

The patient table shows significant differences in non-MASLD vs. MASLD individuals, with the latter suffering more often from diabetes or hypertriglyceridemia. Rather than just stating corrections, subgroup analyses should be performed (accompanied with designated statistical power analyses) to infer the degree to which these conditions contribute to the observations. I.e., major findings stating MASLD-associated changes should hold true in the subgroup of MASLD patients without diabetes/of female sex and so forth (testing for each of the significant differences between groups).

Post-rebuttal update: The authors have performed the requested sub-group analysis and find the gene signatures hold for the non-diabetic sub-cohort, but not the diabetic subgroup. They denote a likely interaction between fibrosis and diabetes, that was not corrected for in the original analysis.

(2) External validation

Additionally, to back up the major GTPase signature findings, it would be desirable to analyze an external dataset of (pre)diabetes patients (other biased groups) for alternations in these genes. It would be important to know if this signature also shows in non-MASLD diabetic patients vs. healthy patients or is a feature specific to MASLD. Also, could the matched metabolic data be used to validate metabolite alterations that would be expected under GTPase-associated protein dysregulation?

Post-rebuttal update: The authors confirm that with the present data, insulin resistance cannot be fully ruled out as a confounder to the GTP-ase related gene signature. They however plan future mouse model experiments to study whether the GTPase-fibrosis signature differs in diabetic vs. non-diabetic conditions.

(3).3D liver spheroid MASH model, Fig. 6D/E

This 3D experiment is technically not an external validation of GTPase-related genes being involved in MASLD, since patient-derived cells may only retain changes that have happened in vivo. To demonstrate that the GTPase expression signature is specifically invoked by fibrosis the LX-2 set up is more convincing, however, the up-regulation of the GTPase-related genes upon fibrosis induction with TGF-beta, in concordance with the patient data, needs to be shown first (qPCR or RNA-seq). Additionally, the description of the 3D model is too uncritical. The maintenance of functional PHHs is a major challenge (PMID: 38750036, PMID: 21953633, PMID: 40240606, PMID: 31023926). It cannot be ruled out that their findings are largely attributable to either 1) the (other present) mesenchymal cells (i.e., mesenchyme-derived cells, such as for example hepatic stellate cells, not to be confused with mesenchymal stem cells, MSCs), or 2) related to potential changes in PHHs in culture, and these limitations need to be stated.

Post-rebuttal update: To address the concern of other cells than hepatocytes contributing to the observed effects in culture, the authors performed TGF-beta treatment in independent mono-cultures (Figure R4): LX-2 and hepatocytes, and the spheroid system. Surprisingly, important genes highlighted in Figure 6E for the spheroid system (RAB6A, ARL4A, RAB27B, DIRAS2) are all absent from this qPCR(?) validation experiment. The authors evaluate instead RAC1, RHOU, VAV1, DOCK2, RAB32. In spheroids, RHOU and RAB32 are down-regulated with TGF-B. In hepatocytes DOCK2 and RAC seemed up-regulated. They find no difference in these genes in LX-2 cells. Surprisingly, ACTA2 expression values are missing for LX-2 cells. Together, it is hard to judge which individual cell type recapitulates the changes observed in patients in this validation experiment, as the major genes called out in Figure 6E are not analyzed.

Unfortunately, the 3D liver spheroid model used (as presented in PMID39605182) lacks important functional validation tests of maintained hepatocyte identity in culture (at the very least Albumin expression and secretion plus CYP3A4 assay). This functional data (acquired at the time point in culture when the RNA expression analysis in 6E was performed) is indispensable prior to stating that mature hepatocytes cause the observed effects.

(4) Novelty / references

Similar studies that also combined liver and blood lipidomics/metabolomics in obese individuals with and without MASLD (e.g. PMID 39731853, 39653777) should be cited. Additionally, it would benefit the quality of the discussion to state how findings in this study add new insights over previous studies, if their findings/insights differ, and if so, why.

Post-rebuttal update: The authors have included the studies into their discussion.

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